

BrainBook

Medium-Fi Prototype

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Value Proposition

A neutral, adaptive AI textbook that gives small hints and asks targeted follow up questions so students build the explanation themselves.



Problem/Solution overview

Current AI tutoring often collapses the learning process into “ask to get answer,” which reduces student agency and skips the step where students externalize and debug their own reasoning.

Solution: We solve this with a neutral, adaptive AI textbook that gives small, digestible hints and then asks targeted follow-up questions so the student actively builds the explanation themselves.

Our Updated Tasks



Simple

The user can answer the three initial prompts to open the textbook.



Moderate

The user can personalize how the AI responds to it by describing how they learn and what tools they like.



Complex

The user can update the textbook with their reasoning based on the questions & explanation they receive.



Task Modifications

- Swapped complexity of Task 2 and Task 3.
- Clarified that reasoning loop is highest cognitive load task.
- Refined task descriptions to focus on user goals, not UI mechanics.

Usability Goal #1



Goal

The user should intuitively know how to interact with the textbook without much help.



Key Measure

The number of times the user misclicks or attempts to do something that they aren't supposed to do.



Progress: *Mostly Achieved*

Core interaction was intuitive, but ending the session and unclear labels caused friction.

- Only 3 usability errors occurred in prior testing, all related to either ending the sessions or labelling



Changes

- Add visible End Session button
 - This button will directly show the user how to end the session
- Rename Favorites → Favorite Learning Tools
 - This label will immediately explain what "Favorites" is referring to

Usability Goal #2

Goal

The user should reduce reasoning errors as the product is used.

Key Measure

The number of times the user makes a mistake in their response, which should decrease over time

Progress: *Partially Achieved*

While diagrams reduced user errors in one experiment, there was no clear reduction of errors in the other experiment.

- Experiment 1: 4 user errors spread throughout, with no clear pattern
- Experiment 2: 3 early errors, but reduction occurred after diagrams were enabled

Changes

- Structure AI feedback with clearer intermediate steps
 - Clearer explanations will naturally improve user understanding
- Make User input a different color
 - This will increase clarity when using the product, which will help with understanding

Sketch Revisions

#1

Old:

BrainBook

Keywords: _____

Questions: _____

Prior Knowledge: _____

Existing

No Open Button

New:

BrainBook

Keywords: ○

Questions: _____

Prior Knowledge: _____

OPEN

EXISTING

extra space to write

Bezel Box for Clarity

open button to direct user

New Existing Page

BrainBook

Topic 1 >

Return

each topic shown in list

Return button

#2

Old:

App User

~ ~ ~

~ ~ ~

No X button

New:

App User

~ ~ ~

~ ~ ~

X button to close app, direct user

User input is a different color for distinction

#3

Old:

Settings

Text Diagram

~ ~ ~

~ ~ ~

Unnecessary Second Page

New:

Settings

Text Diagram

~ ~ ~

~ ~ ~

Single Page for clarity

User input is a different color for clarity

(Overlay)

Sketch Major Changes #1

Redesigned Front Page for Interaction Clarity

Changes Made:

- Added a clear input bezel box to show exactly where users should interact
- Added more writing space for easier input
- Added a visible Open button to remove guessing about how to start
- Added an “Existing” page to complete the UI
 - not part of task flows, but improves system completeness

Why: In GR5, users were unsure how to start and how to move forward. The Open button removes ambiguity and makes the starting action explicit. More writing space supports natural interaction.

Prototype Caveat: In the prototype, the Open button navigates directly to the AI page. In the real product, it would trigger a popup that would instruct the user to physically flip the foldable open.

Sketch Major Changes #1

Assessment of Revision: *Mostly Achieves Usability Goal #1*

- This redesign directly improves Goal #1 (intuitive interaction) by making the start action obvious (clear input box + visible Open button) and giving more space to write.
 - This will reduce “how do I start?” confusion and lower usability errors tied to starting the session.
- This change does not directly address Goal #2 (reduce reasoning errors).
 - However, clearer prompts and more writing space can help users organize their thinking earlier, which may indirectly reduce mistakes during the learning loop.

Future Changes:

- Add a short onboarding tooltip for the 3 starting prompts to better support Goal #1.
- Add example placeholders for each prompt like keywords / question / prior knowledge to improve early input quality and support goal #2.

Sketch Major Changes #2

Improved Session Control & Visual Distinction

Changes Made:

- Added a visible X button to end the session
- Made user input a different color from AI input

Why: In GR5, both users didn't know how to stop using the product. The X button directly fixes that usability error. Color distinction improves clarity between user reasoning and AI feedback.

Prototype Caveat: In the prototype, the X button navigates back to the start page. In the real product, it would trigger a popup instructing users to physically flip the foldable close.

Sketch Major Changes #2

Assessment of Revision: *Partially Achieves* Both Goals

- Adding a visible X button directly improves Goal #1 (intuitive interaction) by removing confusion about how to end the session.
 - This directly eliminates a usability error observed in both GR5 experiments.
 - Making user input a different color from AI input further supports Goal #1 by increasing visual clarity between interaction zones.
- This change also supports Goal #2 (reduce reasoning errors).
 - Clear visual distinction between user reasoning and AI feedback reduces cognitive confusion, which can help users catch mistakes more effectively during the learning loop.

Future Changes:

- Add a confirmation message when ending the session to reinforce clear interaction for Goal #1.
- Strengthen visual hierarchy (e.g., clearer AI step formatting) to further support Goal #2 by helping users track reasoning progression more easily.

Sketch Major Changes #3

Simplified Settings Page

Changes Made:

- Reduced settings to a single clear page
- Removed unnecessary secondary page
- Clarified selectable options

Why: Users previously hesitated or interacted with unclear elements. Simplifying to one page reduces cognitive load and removes meaningless interaction zones.

Sketch Major Changes #3

Assessment of Revision: *Mostly Achieves Usability Goal #1*

- By improving clarity, this design change will reduce usability errors and achieve Goal #1 better
- The changes here don't directly address Usability Goal #2
 - However, increased user clarity and understanding of the settings through distinct coloring will lend itself to better comprehension that will reduce errors over time

Future Changes:

- Add more settings to address different preferences that could improve user experience
 - For example, allow the user to tweak the AI's response style and tools more to get responses that suit the user even better, helping learning

Simple Task Flow

BrainBook

Keywords:

Question:

Prior Knowledge:

Open

Existing

BrainBook

Keywords:

Calculus

Question:

Prior Knowledge:

Open

Existing

BrainBook

Keywords:

Calculus

Question:

How do I integrate for Calculus

I, like basic polynomials and stuff?

Prior Knowledge:

Open

Existing

Click Orange Pill

Click Question Lines

Click Prior Knowledge Lines

Simple Task Flow

BrainBook

Keywords: **Calculus**

Question: How do I integrate for Calculus
I, like basic polynomials and stuff?

Prior Knowledge: I know that there's a
power rule for integrals, like the opposite of the
derivative power rule

Open

Existing

Click Open Button

AI Space

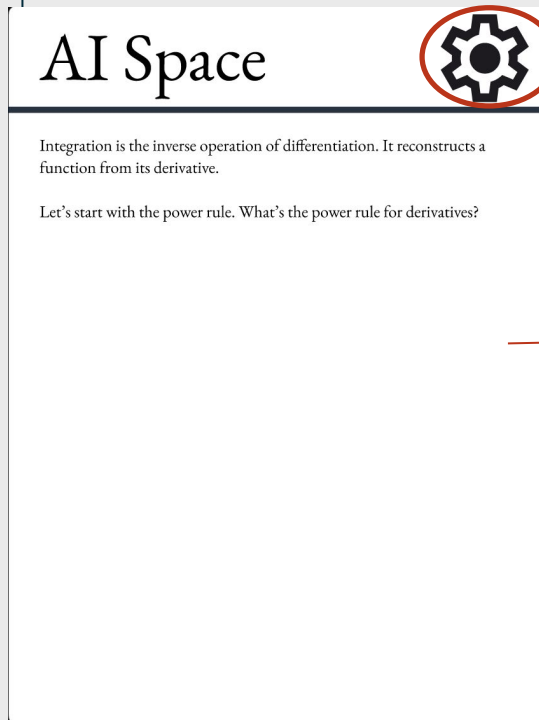


Integration is the inverse operation of differentiation. It reconstructs a function from its derivative.

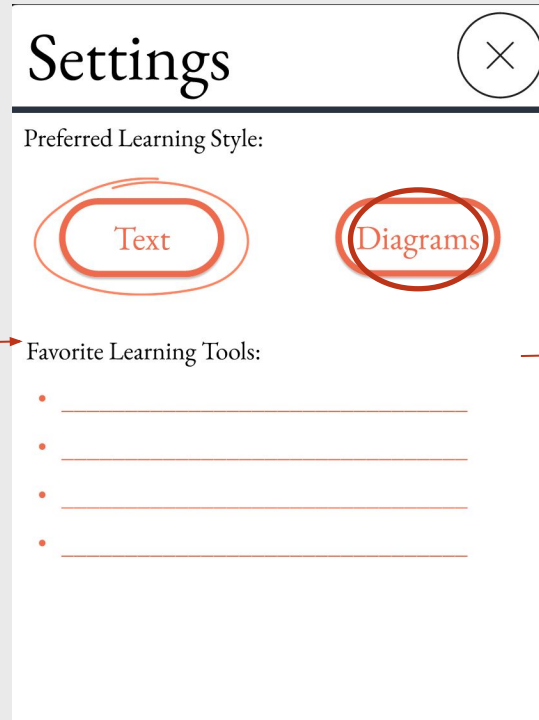
Let's start with the power rule. What's the power rule for derivatives?

Main Page is Opened

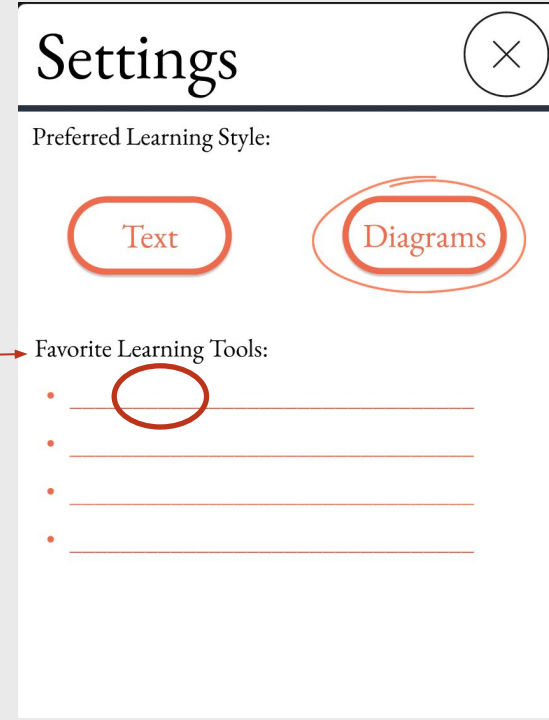
Moderate Task Flow



Click Settings Icon

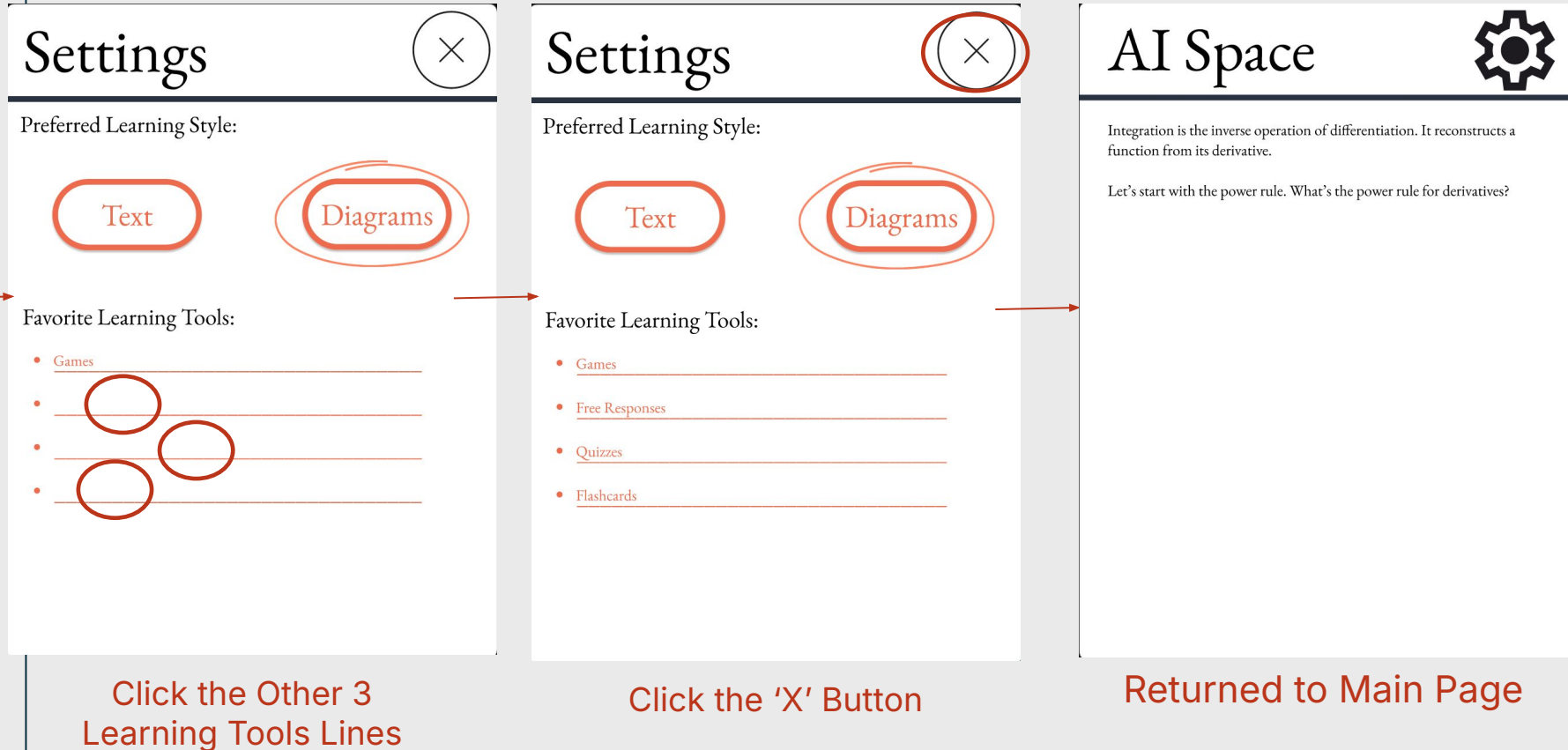


Click Diagrams Button

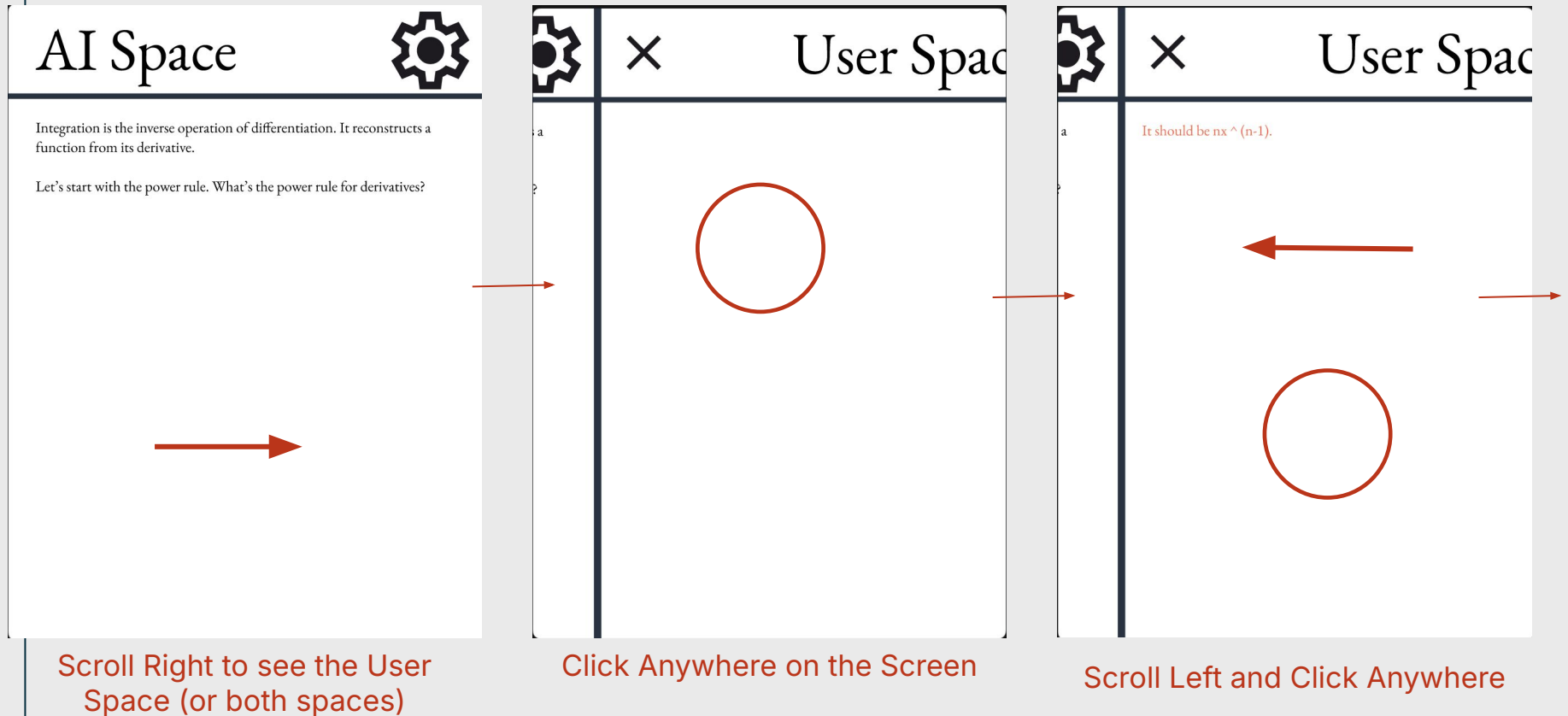


Click Learning Tools Line 1

Moderate Task Flow



Complex Task Flow



Complex Task Flow

AI Space

Integration is the inverse operation of differentiation. It reconstructs a function from its derivative.

Let's start with the power rule. What's the power rule for derivatives?

Right! So what if we wanted to go the other way? For example, let's take $y = x^2$. What's the derivative of that using the formula?

New AI Response

(A) Scroll Right and Click Anywhere

User Space

It should be $nx^{(n-1)}$.

The derivative is $2x$, cus its $(2)x^{(2-1)} = 2x$

New User Response

(B) Scroll Left and Click Anywhere

User Space

of differentiation. It reconstructs a

t's the power rule for derivatives?

ne other way? For example, let's take y using the formula?

do you have to do to $2x$ to get to x^2 ?

y that $(2x) * x/2$ gets us to x^2 . Now,

exact same! If we have nx , we divide by ically, that is just equal to x^2 . But ify?

1, which allow us to generalize for any y x :

ever, because we start with some n , we 1:

$x^{(n+1)}/(n+1)$

Repeat steps (A) and (B) until clicking stops progressing the interaction

Complex Task Flow



et to x^n , and divide by $b+1$
 x^{b+1} , the Power Rule formula is:

x^6 .

arbitrary constant.

differentiation, and that when
0 and goes away.

ve of 2?

us example, all the times we
n number (which we notate as C).

no C! That means that when we
there could be a random
ding C, always.

x^2 ?

olving $-2 + 1$. Remember that we
Then, we get $(x^2-1)/-1$, which

Ohhhhhh, that makes sense! Cus we
 $b+1$ equals n.

Ok, if $b = 6$, $b+1 = 7$. So it should be
What?? Why do we add that? What i

Ok, that should just be 0, right?

A negative number? Woah. Ok, $-2 +$
 $(x^2-3)/-3 + C$, right?

Visual Analysis:
> AI Answer: $-x^2 + C$

Purple: We got -1, not -3. We got tha

Green: The coefficient is -1, not -3. V

Ohhhhhhh, so I just messed up the a
just divide by $b+1$ and add 1 to the e

BrainBook

Keywords:

Question:

Prior Knowledge:

Click 'X' to end the interaction

Initial Page is Reopened

Limitations		
One-screen constraint: The real concept is a foldable with two adjacent screens, but Figma only lets us prototype inside a single upright iPad frame, so we mimic the two-screen layout using horizontal scrolling (the second panel extends past the frame). No free writing/typing: Figma doesn't support real text entry, so we simulate user input with tap/click elements that fill in preset text. No dynamic AI: We can't connect live user responses to an LLM inside Figma, so the AI output is fixed rather than generated in real time. Limited input types: Since users can only click (not circle or draw), preference selection is done through buttons instead of pen-style marking.		

Hard-Coded & Wizard-of-Oz Elements

To simulate the product's functionality, these are hard-coded in the prototype:

- Initial prompts: Keywords, Question, Prior Knowledge
- All AI Space responses
- All User Space filled responses (via click-to-fill)
- All Settings options for Favorite Learning Tools
- All interactions on the Existing Interactions page

Appendix

Prototype Link, README, Extra Sketches

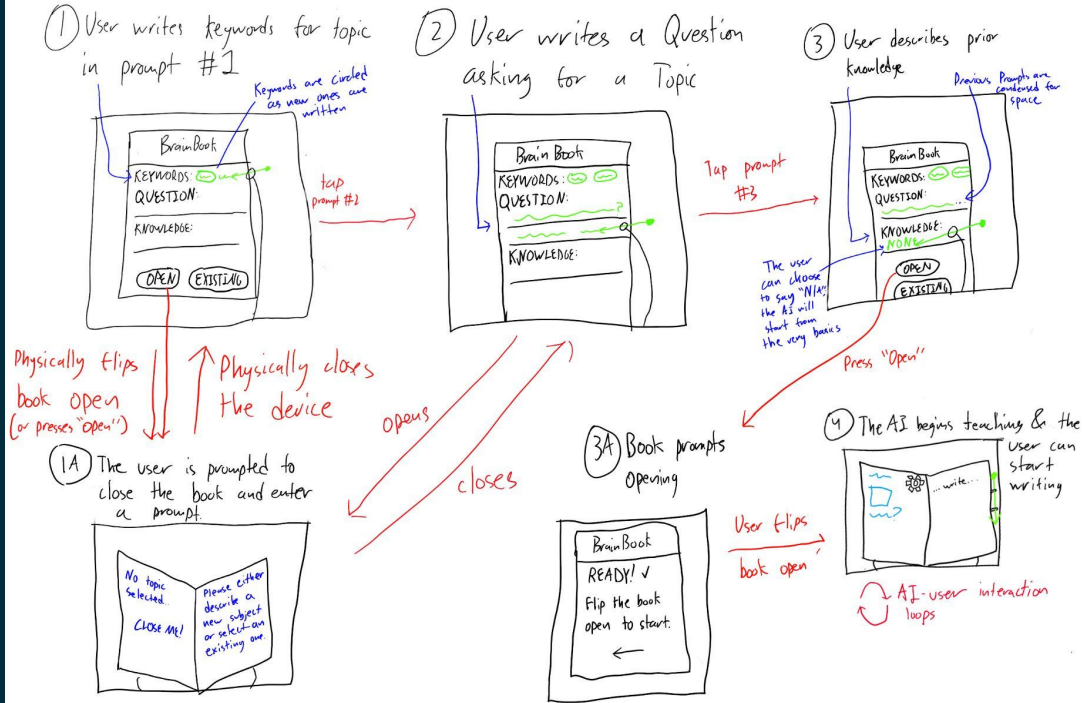
Figma Prototype

README (Google Doc)

README (PDF)

Sketches (Task #1)

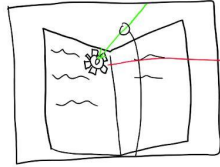
Task #1 Flow



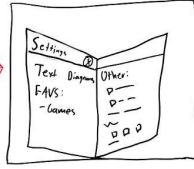
Sketches (Task #2)

Task #2

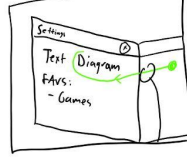
① User taps setting icon



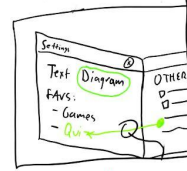
② Textbook shows all settings as-is



② User edits a setting (ie, circling diagram to prefer it)

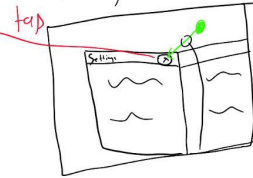


②A User edits another setting

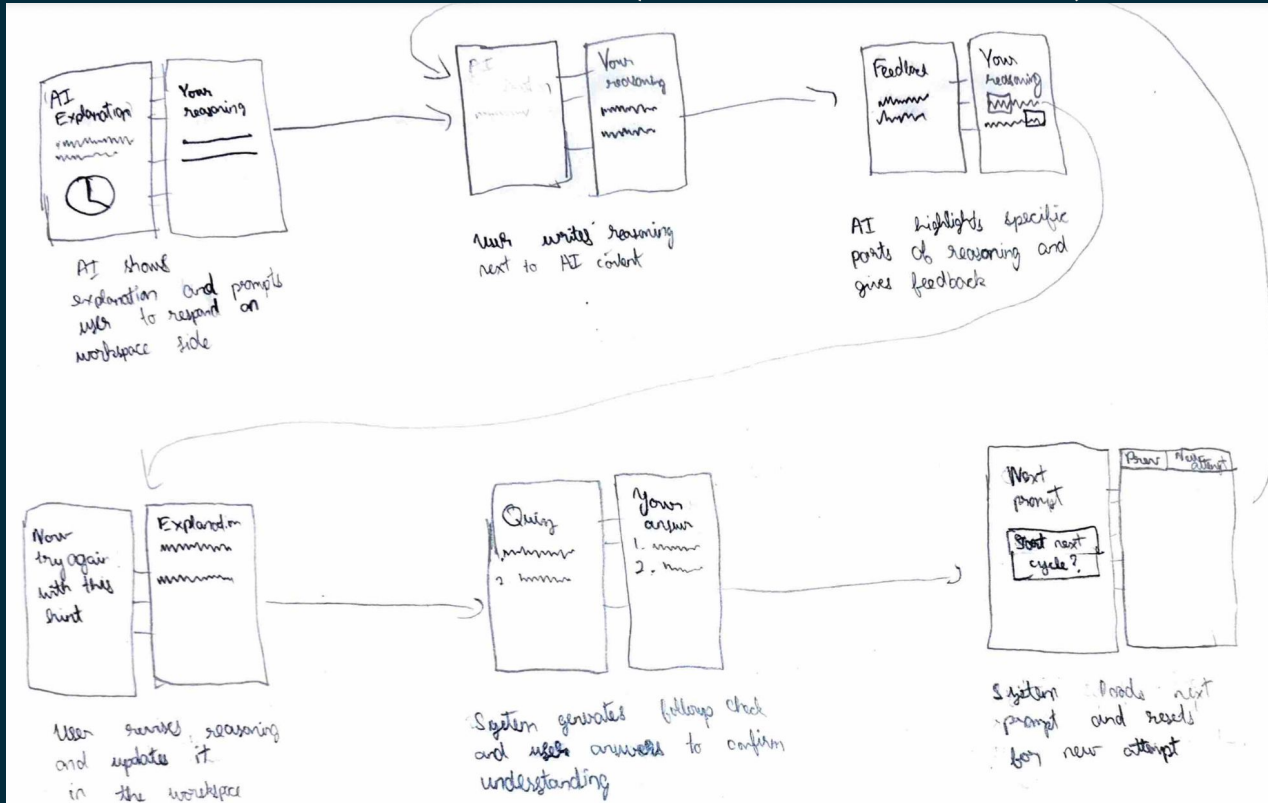


③ User edits an arbitrary number of settings

④ User clicks ① to exit settings menu & continue learning



Sketches (Task #3)





Thank you!